Vanshika Gupta

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Skills

Programming & Computation: Python, PyTorch, C++, SQL
Mathematical & Theoretical Modeling: Probability & Statistics, Econometrics (SVAR, VARX, GARCH)
Core AI/ML: Continual Learning, Test-Time Adaptation, Transformers, LLMs
Scientific Data Engineering: Pandas, NumPy, SciPy, xarray, Geopandas, OpenCV
Advanced Coursework: Natural Language Processing (NLP), Deep Learning, Computer Vision, Artificial Intelligence (AI), Machine Learning, Applied Optimization, Algorithms

Professional Experience

IISER Bhopal | *Visual Data Computing Lab (VisDom)* 01/26 - Ongoing
Research Intern – Continual Test-Time Adaptation (under Prof. Vinod K. Kurmi)

- Researching Continual Test-Time Adaptation (CTTA) under dynamic domain shifts, analyzing state-of-the-art architectures like DPCore to identify novel research gaps.
- Evaluating advanced visual prompting and dynamic coreset mechanisms to prevent catastrophic forgetting and improve real-time model robustness.

IIT Delhi | *DST Centre of Excellence for Climate Information* 05/25 – 07/25
Research Intern – Unsupervised Spatial Fusion (under Prof. Saroj K. Mishra)

- Engineered a Python (xarray, NumPy) pipeline to ingest, re-grid, and harmonize 6 multi-dimensional climate datasets spanning an 18-year period, enabling comparative analysis without an absolute ground truth.
- Designed an unsupervised Generalized TCH algorithm and applied non-linear optimization to evaluate error variances, successfully generating an optimized, SNR-weighted ensemble dataset.
- Isolated regional data discrepancies using a Leave-One-Out variance reduction methodology, automating the generation of spatial uncertainty heatmaps (Geopandas, Matplotlib) to map dataset reliability across the subcontinent.

6thSense | *Freelance* | Remote 12/23 – 04/24

- Developed a real-time 3D pose exercise tracking system utilizing STGCN, CNN, and BiLSTM architectures on a scalable cloud server, optimizing inference concurrency through asynchronous task management (asyncio) and multiprocessing.

Projects

Neural Network Robustness Analysis | *Python, PyTorch, ResNet/ViT, OOD Generalization*

- Evaluated CNN and ViT architectures (ResNet, ConvNeXT) under OOD shifts and image corruptions on CIFAR-10 and ImageNet-100, optimizing validation pipelines and mapping t-SNE decision boundaries to analyze layer sensitivity via internal feature perturbations.

OccFecNet: Robust Facial Expression Recognition | *PyTorch, FaceNet, DenseNet, Knowledge Distillation*

- Engineered OccFecNet via a progressive 4-phase teacher-student distillation curriculum, augmenting a frozen FaceNet backbone with a custom residual spatial attention module and multi-component loss function to extract occlusion-invariant embeddings without inference-time masks, achieving >0.92 mean cosine similarity to the teacher and a >5% accuracy improvement on the heavily occluded FED-RO dataset.

Macroeconomic Time Series Analysis | *SVAR, VARX, Granger Causality, Python*

- Developed robust SVAR and VARX frameworks—utilizing Granger Causality, FEVD, and Impulse Response Functions (IRFs)—to quantify causal shock transmissions, isolate structural breaks, and map endogenous policy responses in multivariate sequence data.

Electricity Consumption Forecasting | *PyTorch, SARIMA-GARCH, Prophet, LSTM*

- Engineered a comparative time series forecasting pipeline evaluating SARIMA-GARCH, Prophet, and custom PyTorch LSTMs, achieving an optimized 2.36% MAPE by explicitly modeling residual volatility clustering and non-linear temporal dependencies.

Video-to-3D Reconstruction (SFM) | *Python, OpenCV*

- Engineered a Structure from Motion pipeline to generate 3D point clouds (`.ply`) from video inputs by extracting SIFT features and optimizing cross-frame mappings via Brute-Force matching and Lowe's Ratio Test.

Education

Indian Institute of Science Education and Research, Bhopal (IISER-B) Bhopal, India
B.S. Engineering Sciences; Major in Data Science Engineering | CPI: 7.73 *2023 – 2027*

Indian Institute of Technology (IIT) Madras Madras, India
Diploma in Programming *2024 – 2027*
Selected Curriculum: System Commands, Modern App Dev, PDSA, Java, Python.

S.V.A. Lalitpur, U.P., India
Senior Secondary (Class 12, CBSE) | 95% *2022*

Leadership & Achievements

- **Fellowships & Ranks**: KVPY Fellow; Qualified JEE Mains & Advanced; All India Rank 1861 in IISER Aptitude Test.
- **Awards**: 2nd Prize for Particle Physics Presentation (IISER-B); Zonal Silver Medalist (SOF Science Olympiad).
- **Leadership**: Student Representative (Centre for Career Development), Mentor (Freshmen Mentorship 2025–26), & Core Committee (EES Club).